AI Code Review Agent

Step 3 – Testing and Deployment Preparation

1. Testing Process

Testing was performed incrementally during implementation to ensure that each component of the system behaves correctly before full workflow integration.

The testing process focused on:

- syntax analysis validation,
- complexity analysis validation,
- intelligent agent workflow testing,
- input handling verification,
- external tool integration testing.

The project uses dedicated testing modules located inside the `tests` directory. Each major component is tested independently to improve reliability and maintainability.

The testing process also verifies that the integrated tools correctly process source code input and return structured outputs without causing unexpected failures.

2. Test Scenarios

Several test scenarios were implemented during development.

Syntax Analysis Testing

The syntax analysis module was tested using both valid and invalid Python source code examples.

The tests verify that:

- valid Python code passes syntax validation,
- invalid Python code correctly produces syntax errors,
- structured outputs are returned consistently.

Complexity Analysis Testing

The complexity analysis module was tested using sample Python functions with conditional logic.

The tests verify that:

- the complexity analysis tool processes source code correctly,
- cyclomatic complexity results are generated,

- structured complexity data is returned.

Intelligent Agent Workflow Testing

The intelligent review agent was tested to verify integration between the agent and analysis tools.

The tests verify that:

- the agent correctly activates analysis tools,
- outputs from multiple tools are collected,
- combined structured results are generated correctly.

3. Deployment Preparation

The project is prepared for controlled local deployment as a Python-based command-line application.

The repository already includes:

- `requirements.txt` for dependency installation,
- modular project structure,
- documentation files,
- testing modules,
- startup scripts.

The system can be installed and executed using the following commands:

```
pip install -r requirements.txt
python main.py
```

The project is designed to be understandable and executable by another user using the provided documentation.

4. Data Conversion and Processing

The system processes Python source code as text input.

Different analysis tools return outputs in different formats. To maintain consistency between components:

- tool outputs are converted into Python dictionaries,
- structured data is normalized,
- the intelligent agent combines processed outputs into unified review results.

This conversion process improves compatibility between modules and simplifies report generation.

5. Current Project Status

At the current stage, the project includes:

- intelligent agent workflow implementation,
- syntax analysis integration,
- style analysis integration,
- complexity analysis integration,
- modular software architecture,
- unit testing structure,
- integration testing structure,
- deployment preparation.

The project is currently in the active implementation and testing phase.